

HARI KRISHNA SURYA PRASAD MERLA

Software Engineer

hmksp12@gmail.com | +1(205) 538 3212 | Birmingham, AL, US | [LinkedIn](#)

SUMMARY

- Around 5 years of experience as a Software Developer specializing in designing, developing, and deploying full-stack applications using Java, React, Spring Boot, and AWS Cloud Services.
- Proficient in building scalable microservices and implementing RESTful APIs using Python and Java, delivering optimized backend solutions for enterprise-grade applications.
- Proficient in front-end development with Node.js, React.js, Next.js, and Tailwind CSS, creating responsive and user-friendly interfaces while employing Python for scripting and data processing.
- Demonstrated expertise in cloud technologies like AWS (EC2, S3, Lambda, and CloudWatch, API Gateway) and DevOps tools such as Jenkins, Docker, and Kubernetes, enabling efficient CI/CD pipelines and reducing deployment times.
- Hands-on knowledge in AI/ML development, including model training, deployment, and integration using TensorFlow, PyTorch, Scikit-learn, and OpenAI APIs, with project experience in NLP, Generative AI (RAG, LangChain), and model interpretability.
- Knowledgeable at designing, responsive front-end applications using modern JavaScript frameworks while leveraging AWS for high-performance, scalable cloud architectures.
- Knowledgeable in architecting and deploying scalable back-end solutions using Java Spring Boot and Python, with a focus on REST API development and seamless integration with AWS cloud services.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, TypeScript, SQL, HTML, JavaScript
- **Frameworks & Libraries:** React, Node.js, Django, Spring Framework (Spring Boot, Spring MVC), jQuery, Flask, Bootstrap, AG Grid, Plotly, Hibernate
- **Databases:** PostgreSQL, Firebase (NoSQL), Oracle APEX, MongoDB, Redis
- **AI & Machine Learning:** Scikit-learn, TensorFlow, PyTorch, XGBoost, LangChain, OpenAI APIs, NLTK, HuggingFace Transformers, BERT, Generative AI, RAG (Retrieval-Augmented Generation), Vector Databases (Pinecone, FAISS), Prompt Engineering
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, CloudWatch, API Gateway, DynamoDB, RDS, IAM, SQS, SNS, CloudFormation), Docker, Kubernetes, Jenkins, Nginx, CI/CD
- **Tools & Platforms:** RESTful APIs, GitHub, Tableau, Dialogflow, ChatGPT, WebGPU, AWS CloudWatch, Firebase

PROFESSIONAL EXPERIENCE

University of Alabama at Birmingham

Data Information Coordinator/ Web Developer

Jun 2024 - Current

Birmingham, AL, US

- Engineered the GeneTerrain visualization web application, employing WebGPU, TypeScript, React, AG Grid, Plotly, and NodeJS, leveraging Python for advanced data visualization and processing to produce over 1 million gene terrains for comprehensive clinical research cancer data analysis.
- Collected GBM clinical cancer data from sources, using Python scripts to reprocess and manage data in an Oracle APEX database; formulated APIs for individual GBM cancer types to serve as input for GeneTerrain software.
- Transformed the Gene Terrain project by utilizing AWS for containerized deployments on EC2, responsive data processing with Lambda, and monitoring via CloudWatch, achieving enhanced performance and scalability.
- Conceptualized Google Firebase for secure authentication, real-time NoSQL data storage, and retrieval, to ensure a scalable and user-centric solution for managing GeneTerrain information.
- Designed RESTful API using Python-based Flask to retrieve and serve gene-gene and gene-pathway data from an Oracle APEX database for the GeneTerrain application, enabling seamless integration and real-time cancer data visualization.
- Implemented API integration using Java Spring Boot and Python, combining the strengths of both technologies to ensure data integrity and high performance.
- Deployed the second version of the Gene Terrain Docker image on AWS EC2 using Nginx, with AWS CloudWatch for monitoring and performance optimization.

Tata Consultancy Services

Software Engineer

Mar 2021 - Jul 2022

Hyderabad, TS, India

- Developed and deployed RESTful APIs using Spring Boot, enabling seamless integration across 10+ microservices and reducing processing latency by 30 seconds per transaction in a high-throughput financial services application.
- Built Web APIs and enterprise modules using Java/J2EE, JDBC, HTML, and the Spring Framework, improving frontend-backend communication and enhancing user experience across 5+ internal tools.
- Managed system performance using AWS CloudWatch and AWS X-Ray, identifying bottlenecks and reducing application response time by 150 milliseconds, improving reliability for critical client-facing platforms.
- Engineered backend components with Hibernate, creating 50+ HQL queries, mapping files, and data access layers to ensure efficient database interactions and transactional consistency.
- Contributed to data-driven automation initiatives by developing Python-based scripts for log parsing, error tracking, and

backend testing, reducing manual debugging effort by 40%.

- Collaborated with internal R&D teams to support machine learning model deployment workflows, integrating Python-based ML models into Java microservices via REST APIs for real-time predictions in customer analytics modules.
- Participated in early-stage prototyping of AI-based chatbot functionality, integrating OpenAI API with internal ticketing platforms to automate Tier 1 support queries and reduce human intervention.
- Contributed to data preprocessing tasks for ML pipelines using Pandas and NumPy, ensuring clean and structured input for internal fraud detection and pattern recognition models.

Assistant Software Engineer

Sep 2019 - Mar 2021

- Developed a chatbot using Python, Django, and Dialogflow in collaboration with the Business Intelligence team to automate HR-related queries, benefiting 50,000+ employees at TCS.
- Architected feedback mechanisms on user responses for real-time data collection for insightful user behavior analysis, utilizing PostgreSQL db leading to a significant improvement in product usage.
- Created a Python automation tool that enables seamless conversion between Excel data and Dialogflow bots, drastically reducing the bot creation time by 90%, and facilitating bot creation within a minute.
- Co-ordinated agile sprints, participated in sprint reviews and retrospectives to improve team performance, and implemented real-time feedback mechanisms and Python automation tools.

Trinity Technolabs

Aug 2018 - Aug 2019

Software Engineer

India

- Collaborated on developing enterprise-level applications using Django, contributing to modular components that processed over 10,000 transactions daily in line with industry standards for scalability and reliability.
- Integrated RESTful APIs using Python to facilitate seamless communication between microservices, handling over 25,000 API requests daily.
- Guided in designing and implementing MongoDB solutions for managing unstructured financial data, including audit records, with secure storage and retrieval for datasets exceeding 100 GB.
- Aided the Redis configuration for caching frequently accessed data, improving database query latency from 120ms to 75ms during peak traffic, and optimizing application performance.
- Administration of GitHub repositories, reviewed pull requests and CI/CD pipelines for automated testing, and streamlined deployment workflows.
- Automated CI/CD pipelines using Jenkins, containerized financial services with Docker, reducing deployment time from 20 minutes to 10 minutes.
- Designed secure RESTful APIs to integrate 3 third-party financial APIs, enhancing interoperability and enabling real-time data exchange between microservices.
- Integrated Tableau for real-time financial dashboards, ensuring responsive data visualization across applications.
- Built React.js interfaces for financial tools, enabling customers to manage real-time account operations and reports with seamless navigation.
- Developed front-end components using Numpy, ensuring interactive user interfaces and improved data visualization.
- Unified cloud services using AWS S3 for file storage and AWS EC2 for hosting, ensuring minimal downtime and cost efficiency.

EDUCATION

University of Alabama at Birmingham

Aug 2022 - May 2024

Master's Degree in Data Science

Birmingham, AL, US

KL University

Jul 2015 - May 2019

Bachelor's Degree in Computer Science Engineering

Guntur, TS, India

PROJECTS

- **Game Zone:** Constructed Game Zone, a video game discovery web app, using React 18, TypeScript, Chakra UI, and Bootstrap. The project features a modern UI, state management, and integration with a backend for content display ([Link](#)).
- **Clinical Data Insights using RAG Model:** Engineered a RAG model with Flask, React, LangChain, Firebase, and OpenAI to analyze patient records, provide summaries, detect anomalies, and identify key health indicators using advanced AI techniques ([Link](#)).
- **Comedian ChatBot:** Created Comedian AI Assistant for humor generation using Langchain, OpenAI, Streamlit, and HuggingFace Spaces, hosted on render cloud for accessibility([Link](#)).
- **SavingsGuru:** Devised a Flask web application in a UA 24-hour hackathon, facilitating user analysis and providing solution to user's spending habits. Analyzed Python, Flask, OpenAI, Plotly, and Kaggle for comprehensive analysis and visualization capabilities ([Link](#)).
- **MovieHub:** A website enabling users to retrieve comprehensive movie information, employing HTML, CSS, JavaScript, REST API, and React technologies. Please find the GitHub code ([Link](#)).
- **Language Prediction:** Built a project that predict language type based on given input. Tools used are Python, JavaScript, Django, Numpy, Pandas, Natural Language Processing, TFIDF, and Sklearn. Please find the code ([Link](#)).
- **GameGuruBot:** Tuned Hugging Face openai-gpt model using a gaming dataset to enhance answer gaming-related questions; leveraged pandas, NLTK, and transformers libraries to preprocess and fine-tune the model.